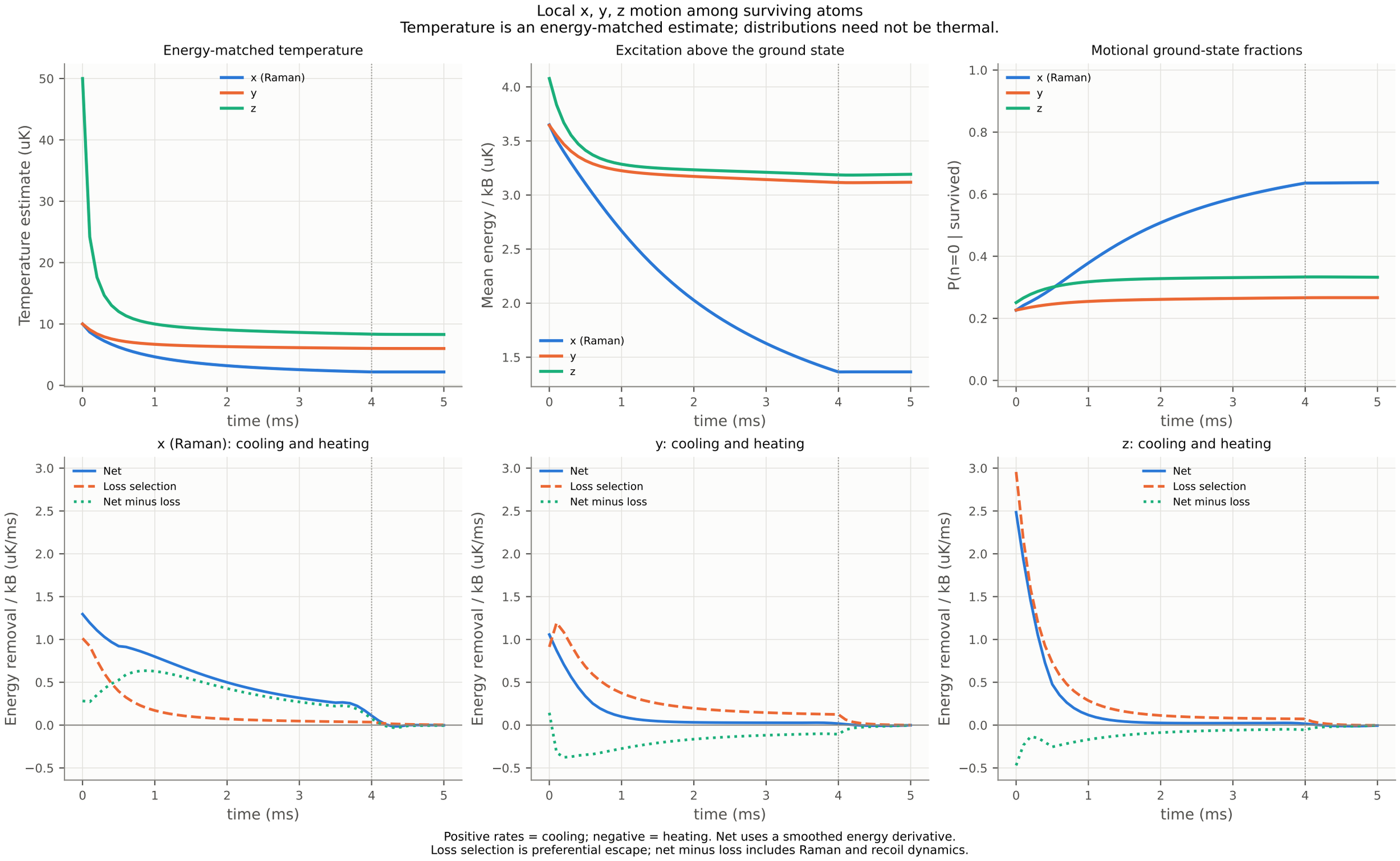
Local x, y, z motion among surviving atoms
Temperature is an energy-matched estimate; distributions need not be thermal.
Energy-matched temperature
Temperature estimate (uK)
x (Raman)
y
z
time (ms)
Excitation above the ground state
Mean energy / kB (uK)
x (Raman)
y
z
time (ms)
Motional ground-state fractions
P(n=0 | survived)
x (Raman)
y
z
time (ms)
x (Raman): cooling and heating
Energy removal / kB (uK/ms)
Net
Loss selection
Net minus loss
time (ms)
y: cooling and heating
Energy removal / kB (uK/ms)
Net
Loss selection
Net minus loss
time (ms)
z: cooling and heating
Energy removal / kB (uK/ms)
Net
Loss selection
Net minus loss
time (ms)
Positive rates = cooling; negative = heating. Net uses a smoothed energy derivative.
Loss selection is preferential escape; net minus loss includes Raman and recoil dynamics.